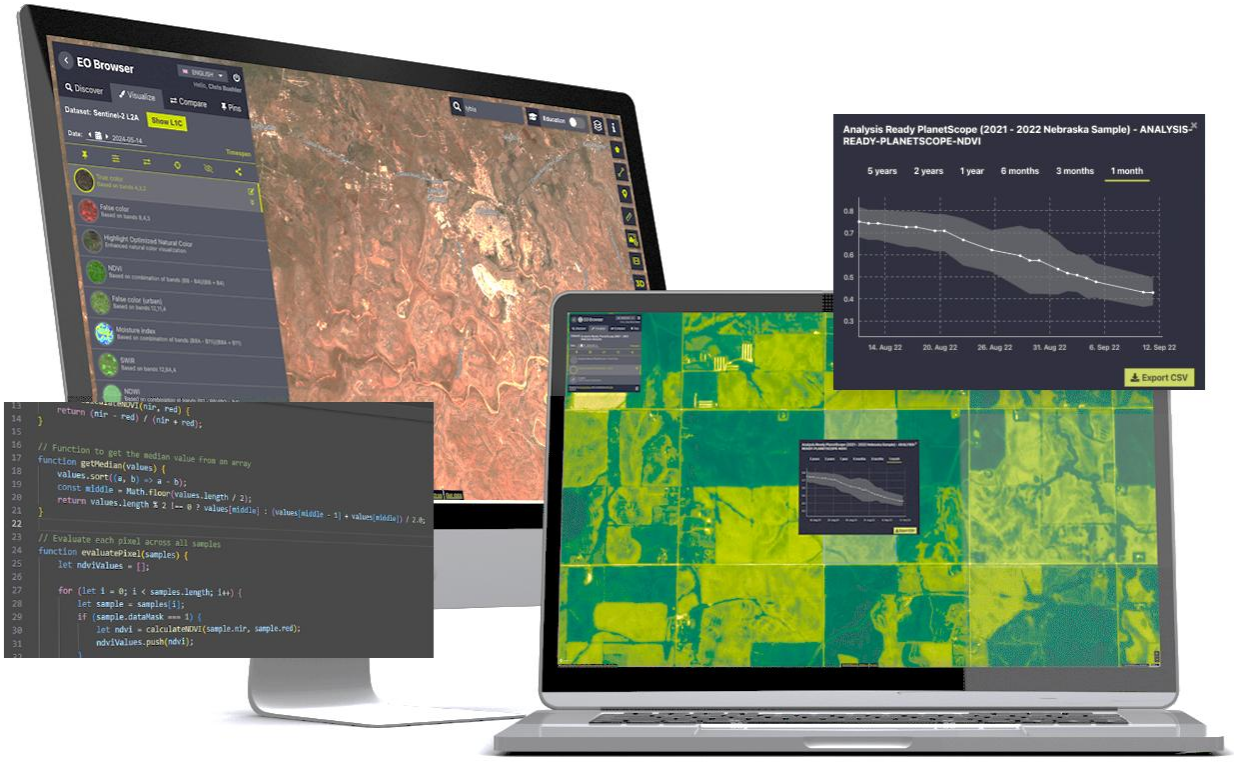
Technical Proposal For Development of Integrated Frameworks and Digital Solutions for FLR Monitoring, Evaluation, and Reporting under RECAP4NDC project



For

giz Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH

Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH

Submitted by

GISFY Building a smart world

GISFY Private Limited

DSL Abacus IT Park, Industrial Development Area Uppal,
Ranga Reddy Dist., Hyderabad, Telangana, 500039, India

Technical Submission Form

GISFY PRIVATE LIMITED
CIN: U74999UP2016PTC087694
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Hyderabad
30th April 2025
Tel No. +91 9971777963
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To,
The Head of Contracts and Procurement
GIZ GmbH, GDCO Office
New Delhi – 110 057

Subject: Proposal in response to Tender No. 83485307

Dear Sir/ Madam,

I/We the undersigned hereby offer to execute the scope of work and accordingly submit our offer in full compliance with terms & conditions of the bid.

The bid is being submitted as per the instructions mentioned in the tender documents.

Legal Name of Entity	GISFY Private Limited
Name of the Authorized Signatory	Reshma Parveen Shaik
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CIN Number/Registration Number	1. Uttar Pradesh: U74999UP2016PTC087694 2. Telangana: U74999TG2016PTC139570
Name of Directors with DIN/PAN	Reshma Parveen Shaik - 07621934 Azghar Hussain Mohammed - 07621930

For and on behalf of GISFY Private Limited.

Ms. Reshma Paveen Shaik
CEO, GISFY Private Limited

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How GISFY meets the technical requirements as per TOR

Fulfilment of technical-methodological concept requirements

S. No.	Technological-Methodological Concept	How we meet the requirement
1	Scalable and Modular System Architecture	We employ a microservices-based architecture, ensuring scalability, modular expansion, and interoperability with existing forest management systems.
2	Integration of Sentinel Hub APIs for Real-Time Satellite Imagery	Our team has extensive experience in integrating Sentinel Hub APIs for real-time satellite imagery retrieval, vegetation index analysis, and automated deforestation detection.
3	Development of Cross-Platform Mobile Applications	We develop mobile applications using Flutter for cross-platform (iOS & Android) compatibility, ensuring seamless field data collection with offline functionality.
4	Implementation of GIS-Based Decision Support System	We utilize OpenLayers, PostGIS, and GeoServer to build a GIS-based decision support system with interactive mapping and data analytics.
5	Real-Time Data Synchronization Between Mobile and Web Systems	We ensure seamless real-time data synchronization using Firebase and RESTful APIs, enabling immediate access to field data from web platforms.
6	User-Friendly UI/UX for Field and Administrative Users	Our UI/UX team follows human-centered design principles, ensuring intuitive navigation for both technical and non-technical users.
7	Role-Based Access Control and Data Security Measures	We implement secure authentication (OAuth 2.0, JWT) and data encryption to protect sensitive forest management data.
8	Automated Forest Cover Change Detection Using NDVI/NDWI	Our team specializes in remote sensing and machine learning models to automate NDVI/NDWI-based forest cover change detection and monitoring.
9	Cloud-Based Deployment for High Performance and Reliability	We deploy the system on AWS and Azure cloud platforms with auto-scaling, load balancing, and disaster recovery features.
10	Comprehensive Training and Knowledge Transfer Strategy	We provide structured training sessions, user manuals, and on-demand support to ensure successful adoption and sustainability of the platform.

Organizational Details

GISFY Private Limited is a is a globally recognized leader in digital transformation, geospatial intelligence, climate risk analysis, and conflict early warning systems. With a strong foundation in ICT system integration, data-driven decision-making, and AI-powered analytics, the organization specializes in developing scalable technology solutions for government agencies, international organizations, and humanitarian entities. Our expertise spans public sector digitalization, early warning platforms, disaster risk management, and predictive modeling, making us a trusted partner in enhancing resilience, security, and governance worldwide. We are recognized as a start-up by the Department of Industrial Promotion and Policy; and have worked with renowned organizations like Nanyang Technological University (Singapore), University of West Indies, Innovative Hydrology, and World Business Council for Sustainable Development (WBCSD), Belize Social Investment Fund, and various Government departments across the world.

We work across the world and provide services and solutions in the following areas of work.

- Agricultural and Forestry Applications Development
- Natural Resources Management
- GIS and ICT Platforms (web and mobile)
- Sentinel Hub API Integration
- Urban Development
- Hydromet and Conflict Early Warning Systems
- Geospatial Intelligence & GIS-Based Risk Mapping
- Disaster Risk Management & Climate Resilience
- Android Mobile App Development
- Geospatial mapping and Data Development
- Documentation & Knowledge Management

The following are further details and documentation proof of our organization

Item	Description
Name of the organization	GISFY PRIVATE LIMITED CIN No.: U74999UP2016PTC087694 PAN No.: AAGCG5903P GSTIN: 36AAGCG5903P1Z6
Type of the organization	Private Limited
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Contact Details	Azghar Hussain Mohammed, Founder & CTO Tel: +91-9971777963 Email: azghar@gisfy.co.in; skype: yashikakki.

Management Structure

Key Personnel	Brief Profile
Dr. G. Sailu 	Dr. G. Sailu is a seasoned Project Manager with over 18 years of experience in leading complex, multi-stakeholder assignments in the domains of environmental management, forestry, and sustainable landscape restoration. He holds a Ph.D. in Environmental Science from Osmania University and an M.Sc. in Biotechnology from Periyar University. Dr. Sailu has successfully overseen the design and implementation of large-scale development projects funded by multilateral agencies, with a focus on planning, monitoring, team coordination, and client engagement. In his role as Project Manager for the RECAP4NDC assignment, Dr. Sailu will provide strategic leadership and full-cycle management oversight for the development of integrated Monitoring, Evaluation, and Reporting (MER) frameworks and associated digital tools for Forest Landscape Restoration (FLR). He will be responsible for managing work planning, expert coordination, stakeholder consultations, compliance with deliverables, and quality assurance across both Work Package 1 and Work Package 2. Dr. Sailu's experience includes managing interdisciplinary teams across diverse geographies, including India, Bhutan, and Kenya, ensuring timely delivery of project outputs while maintaining alignment with technical standards and institutional requirements. He brings a strong background in coordinating with national and state-level government institutions such as MoEFCC, FSI, and State Forest Departments, ensuring seamless integration of project activities with government priorities. He has led national-level assignments involving restoration planning, spatial data integration, and institutional capacity building. His ability to manage cross-functional teams including GIS experts, software developers, remote sensing analysts, and training specialists makes him well-suited to deliver the RECAP4NDC project objectives within the stipulated timeline and budget. Dr. Sailu will ensure adherence to the TOR-specified methodology, guide the implementation of stakeholder engagement and validation processes, and oversee the piloting of MER tools in Gujarat, Maharashtra, Uttarakhand, and Delhi NCR. His experience managing donor-funded projects aligns closely with GIZ's expectations on performance reporting, visibility, and adaptive planning.

	In addition to his project management responsibilities, Dr. Sailu will contribute to capacity-building efforts, coordinating the delivery of training modules, user manuals, and field implementation protocols. His leadership will ensure that the MER framework and digital solutions are not only technically sound but also practical, user-friendly, and sustainable for long-term institutional adoption. Dr. Sailu's proven project management track record, domain familiarity, and results-driven approach will be instrumental in delivering a high-impact, scalable MER system under the RECAP4NDC initiative.
Mr. Mohammad Moonis	Mr. Mohammad Moonis is a highly skilled Data Scientist and environmental analytics expert with a M. Phil. Natural Resource Management from the Indian Institute of Forest Management, Bhopal, and a M. Sc. Environmental Science from Jawaharlal Nehru University, New Delhi. With over 17 years of experience in data-driven environmental modeling, machine learning, and geospatial data analytics, Dr. Moonis brings a unique blend of domain expertise and advanced technical proficiency to support evidence-based forest management and landscape restoration. Dr. Moonis specializes in the application of statistical modeling, predictive analytics, and AI-based algorithms for environmental monitoring, land-use change analysis, and ecological risk assessment. His expertise will be critical to the development of intelligent features within the geospatial dashboard and mobile applications under the RCAP4NDC Gujarat project, ensuring real-time data interpretation, trend forecasting, and spatial decision support. Having worked extensively with international organizations such as FAO, UNEP, and GIZ, Dr. Moonis has led the design of machine learning models for deforestation prediction, carbon stock estimation, and afforestation impact evaluation. He is proficient in tools such as R, Python, Google Earth Engine, TensorFlow, and PostgreSQL/PostGIS, enabling him to build scalable, automated data pipelines and high-performing analytical dashboards. His work supports seamless integration of satellite imagery, mobile-based field data, and forest inventory datasets to produce actionable insights for planners and policymakers. Within this project, Dr. Moonis will be responsible for designing the data architecture and analytical models that underpin key functionalities, such as forest cover change detection (using NDVI/NDWI), patrolling pattern analysis, and spatial validation of working plan coupes. He will also lead the development of

	real-time alert systems and performance indicators that enhance the Department's ability to monitor restoration efforts and make data-backed interventions. Dr. Moonis has previously contributed to national-level decision-support tools and WebGIS platforms for forest degradation monitoring, agroforestry planning, and climate vulnerability mapping. His experience in combining structured and unstructured datasets, performing multi-temporal satellite image analysis, and building AI-enabled classification models will directly support the development of an intelligent and responsive geospatial system for the Forest and Environment Department, Government of Gujarat. In addition to his technical contributions, Dr. Moonis is a strong advocate of open data standards, reproducible research, and institutional capacity building. He will assist in creating user-friendly data visualization interfaces and analytical dashboards for use by field officers, GIS teams, and policy stakeholders. He will also lead sessions on data literacy and analytics interpretation as part of the project's knowledge management and training components. With a proven track record of using data science to drive conservation outcomes and policy transformation, Dr. Moonis will play a pivotal role in ensuring that the RCAP4NDC digital monitoring tools are robust, insightful, and future-ready—empowering Gujarat's Forest administration with the analytical capabilities needed for sustainable and adaptive forest landscape restoration.
Dr. Jayanthi Nagakumar 	Dr. Jayanthi Nagakumar is a distinguished Ecosystem Modeller and landscape ecologist with a Ph. D (Environmental Planning) from University of Mysore, and a M.Sc. (Botany) from University of Mysore. With over 36 years of experience in ecosystem simulation, spatial habitat modeling, and land degradation assessment, Dr. Jayanthi brings deep domain knowledge and advanced modelling expertise to support sustainable forest management and Forest Landscape Restoration (FLR) under climate change scenarios. Dr. Jayanthi has worked extensively at the interface of ecology, remote sensing, and spatial analytics, using data-driven models to simulate ecological processes, assess forest resilience, and support biodiversity conservation planning. Her contributions will be central to the RCAP4NDC Gujarat project, where she will develop ecosystem simulation models and restoration suitability maps, ensuring that mobile and dashboard tools are rooted in ecological intelligence and long-term environmental planning.

	With a career spanning multi-country projects in India, Dr. Jayanthi has applied ecosystem modeling techniques such as species distribution modeling (SDM), landscape connectivity analysis, and carbon sequestration forecasting to inform conservation strategies, afforestation programs, and natural resource management. She is proficient in platforms like InVEST, MaxEnt, R, GAMS, and ArcGIS ModelBuilder, allowing her to integrate ecological datasets with geospatial workflows for scenario analysis and restoration prioritization. As part of this project, Dr. Jayanthi will lead the design and implementation of spatial ecosystem models to assess forest health, land-use change dynamics, and restoration potential across Gujarat's forest landscapes. Her work will support the identification of ecologically sensitive zones, validate land cover classifications, and contribute to the development of decision-support layers within the geospatial dashboard enabling forest managers to visualize trade-offs and co-benefits of restoration actions. Dr. Jayanthi has previously contributed to the development of degradation vulnerability maps, ecosystem services quantification tools, and climate-resilient land-use strategies, making her expertise invaluable for building an integrated decision-making platform under the RCAP4NDC framework. Her work aligns with global standards and initiatives such as the Bonn Challenge, REDD+, and UNCCD Land Degradation Neutrality goals, ensuring policy and institutional relevance. In addition to his modelling work, Dr. Jayanthi is deeply involved in capacity building and applied research translation. She will support the creation of ecological training modules, conduct technical workshops for forest officers and GIS analysts, and guide the Forest Department in interpreting ecosystem indicators and model outputs for planning and monitoring purposes. With a rare combination of ecological insight, spatial modeling expertise, and hands-on experience in forestry and restoration contexts, Dr. Jayanthi will play a critical role in ensuring that the Gujarat Forest Monitoring platform is not only technologically advanced but also ecologically grounded and restoration-oriented contributing directly to Gujarat's vision of sustainable and climate-resilient forest landscapes.
Mr. Azghar Hussain Mohammed	Mr. Azghar Hussain Mohammed is a highly accomplished Software Engineer and enterprise systems architect with a Bachelor's degree in Computer Science Engineering from Kakatiya University, Andhra Pradesh, and over 22 years of experience in software development, platform architecture, and large-scale system integration across environmental, agricultural, and e-governance domains. His deep expertise in

full-stack development, secure API design, and high-performance digital platform engineering makes him exceptionally qualified to lead the software development components of the RCAP4NDC Gujarat Forest Monitoring Project.

Mr. Hussain has led the development of complex digital solutions for government and donor-funded projects, integrating WebGIS, mobile applications, and cloud infrastructure to address challenges in natural resource management, forestry, and climate adaptation. His core competencies include end-to-end software architecture, responsive mobile application design, backend service orchestration, and platform security compliance, with proficiency in technologies such as Microsoft ASP.NET, PHP, Python, Angular, React, and C#. He also brings strong experience in PostgreSQL, MongoDB, MySQL, and real-time data streaming from field-based IoT/GIS sources.

For this project, Mr. Hussain will play a central role in the design, development, and deployment of the mobile applications and the centralized geospatial dashboard, ensuring seamless functionality, real-time synchronization, and cross-platform scalability. His contributions will include building robust backend systems for forest cover change detection, patrol reporting, and coupe ground-truthing workflows ensuring that field data is processed securely and visualized instantly within the decision-support dashboard.

Mr. Hussain has previously developed cloud-based platforms for forest monitoring, seed supply chain tracking, e-marketplaces, and climate-vulnerable region assessments, supporting organizations such as the World Bank, UNDP, and state forest departments.

He is also adept at integrating geospatial processing tools (GeoServer, Leaflet, Mapbox) and real-time analytics modules, allowing the Gujarat Forest Department to monitor satellite data trends, generate reports, and respond to deforestation alerts with precision. His contributions will ensure that the digital tools are not only responsive and user-friendly but also compliant with data protection laws, ensuring secure handling of sensitive environmental and geolocation data.

Beyond development, Mr. Hussain plays a strong mentoring and support role facilitating developer team coordination, DevOps pipeline automation, and quality assurance testing to ensure agile delivery and high system reliability. He will also support capacity building by creating technical documentation,

	user training materials, and system maintenance protocols tailored to the needs of the Forest Department's IT teams. With a proven track record in delivering scalable and secure digital solutions for complex environmental systems, Mr. Azghar Hussain Mohammed will be instrumental in ensuring that the Gujarat RCAP4NDC platform is technically sound, fully interoperable, and capable of supporting real-time forest management and FLR monitoring at scale.
Mr. Durga Ajay Babu 	Mr. Ajay Babu is a highly skilled UI/UX designer with extensive experience in creating user-friendly and visually engaging digital interfaces for web and mobile applications. His expertise lies in user-centered design, accessibility, and interactive prototyping, ensuring seamless navigation and an intuitive user experience. He specializes in wireframing, prototyping, and front-end optimization, leveraging industry-standard tools such as Adobe XD, Figma, Sketch, and Axure RP. His technical proficiency in HTML, CSS, JavaScript, and Bootstrap enables him to develop pixel-perfect, responsive designs tailored to meet diverse user needs. He has played a crucial role in designing digital solutions for projects such as TIGRAM – Timber Traceability, Agri Entrepreneur Growth Foundation (AEGF), and NTR Smart Township, where he ensured optimal user experience, interactive visualization, and accessibility compliance. His work spans India, Africa, Cambodia, Nigeria, Belize, Vietnam, Singapore, Tajikistan, Kenya, Ethiopia, Bermuda, and the USA, reflecting his ability to adapt to global design trends and usability standards. His ability to translate complex user requirements into engaging, functional digital interfaces makes him an invaluable asset in enhancing the usability and effectiveness of digital platforms.
Mr. Pranay Kumar 	Mr. Pranay Kumar is an accomplished Geospatial Expert with a Master's Diploma in GIS & Remote Sensing from Jadavpur University, Kolkata, and a Master's in Geography from Patna University, Bihar. His strong academic background in geospatial science, cartography, and satellite image processing has provided him with the expertise to develop GIS-based solutions for environmental monitoring, land-use planning, and sustainable resource management. With an extensive understanding of spatial analysis, remote sensing applications, and geospatial data visualization, Pranay has played a pivotal role in integrating GIS technology with decision-support systems. His deep knowledge of cartographic principles, GIS modeling, and cloud-based geospatial data processing has allowed him to contribute effectively to complex GIS projects.

	With over 13 years of experience in GIS development, geospatial database management, and spatial decision-making, Pranay has worked on large-scale GIS implementations in India, Malawi, and Afghanistan. His expertise includes developing WebGIS applications, designing geospatial data infrastructure, and integrating remote sensing technologies for land-use analysis. He has contributed to various GIS-driven projects in forestry, agriculture, and disaster risk management, leveraging geospatial intelligence to support policy formulation and environmental sustainability initiatives. His work has been instrumental in building GIS portals, developing geospatial analytics models, and designing interactive mapping platforms for real-time decision-making. Pranay has been actively involved in conducting spatial risk assessments, mapping ecosystem vulnerabilities, and integrating climate adaptation strategies into GIS frameworks. His hands-on experience in WebGIS development, GIS database architecture, and cloud-based spatial data processing aligns seamlessly with this project's focus on forest monitoring, real-time satellite imagery integration, and mobile-based GIS applications. He is highly proficient in ArcGIS, QGIS, PostGIS, GeoServer, and OpenLayers, making him well-equipped to develop and deploy WebGIS applications for large-scale environmental projects. His expertise in Sentinel-2 and Landsat satellite image analysis enables him to extract valuable insights for forest cover change detection, vegetation health assessment, and afforestation monitoring. He has extensive experience in geospatial database design, spatial query optimization, and GIS-based predictive modeling, ensuring that the data-driven insights generated from GIS applications contribute effectively to conservation planning and forest management. His technical proficiency in developing spatial models, integrating AI-driven remote sensing analytics, and automating GIS workflows ensures that geospatial applications are scalable, efficient, and user-friendly. Throughout his career, Pranay has worked on numerous projects that utilize GIS technology for environmental monitoring, land degradation assessment, and sustainable forestry management. His work in developing vulnerability maps, land-use classification models, and geospatial dashboards for climate adaptation strategies demonstrates his ability to apply GIS in addressing environmental challenges. His experience in Sentinel Hub API integration for real-time remote sensing applications ensures that the forest monitoring platform

	will have highly accurate, up-to-date geospatial intelligence for conservation efforts. His expertise will be crucial in ensuring that the GIS system effectively supports data-driven decision-making, enabling policymakers and conservationists to assess afforestation efforts, detect deforestation trends, and optimize forest resource management strategies. With his strong background in geospatial science, GIS application development, and environmental analytics, Pranay will play a vital role in delivering a scalable, efficient, and impactful GIS solution for Gujarat's Forest conservation initiatives.
Mr. Kishan Kumar 	Mr. Kishan Kumar is an experienced full-stack web developer with over 8 years of expertise in developing scalable, high-performance web applications. His technical proficiency includes ASP.NET, PHP, Angular, ReactJS, Python, and JavaScript frameworks, along with extensive knowledge of database management systems such as MongoDB, PostgreSQL, and SQL Server. His expertise lies in developing dynamic, secure, and data-driven web platforms, integrating real-time analytics, cloud-based solutions, and robust API frameworks. He has contributed significantly to projects such as TGP, Lekha – Money Monitoring, and NTR Smart Township, where he played a key role in integrating secure payment gateways, WebGIS functionalities, and responsive data visualization tools. Having worked across India, Africa, Vietnam, Singapore, and the USA, Mr. Kishan has successfully developed and deployed enterprise-grade web solutions tailored to meet the needs of government agencies, private enterprises, and development organizations. His knowledge of agile methodologies, cloud security, and front-end optimization makes him a vital resource for modern, scalable, and high-performance web development.
Ms. Pankaja Chichagandi 	Ms. Pankaja Chichagandi is a seasoned full-stack developer with over 7 years of experience in software and web development, specializing in GIS-based applications, cloud computing, and fintech platforms. She has a strong background in C#, Java, PHP, Laravel, Drupal, and Python, coupled with expertise in GIS technologies such as ArcGIS, OpenLayers, and GeoServer. She has played a pivotal role in projects such as TIGRAM – Timber Management and Traceability, Agri Entrepreneur Growth Foundation (AEGF), and Van System, where she ensured seamless system integration, database optimization, and performance-driven application development. Her ability to

	develop intuitive, responsive, and scalable web solutions has contributed to the success of multiple data-driven projects in agriculture, forestry, and financial inclusion. Her experience across India, Africa, Singapore, and the USA highlights her adaptability to global technology trends and business requirements. With her expertise in web application security, cloud architecture, and API integrations, Ms. Pankaja is a key asset in delivering robust, secure, and innovative digital solutions.
Ms. Girija Priyadarsini Nayak 	Girija Priyadarshini Nayak is a highly proficient Web Developer with a Master's in Computer Applications (MCA) from Khallikote University, Odisha, India. With a strong academic foundation in software development, database management, and front-end/back-end integration, she has developed expertise in building scalable, secure, and high-performance web applications. Her education has equipped her with a deep understanding of programming languages, cloud-based deployments, and UI/UX design principles, ensuring that she can deliver seamless and interactive digital experiences. Her strong analytical skills, combined with her proficiency in modern web technologies, security protocols, and application optimization, make her a critical asset for complex IT solutions, especially for geospatial platforms and decision-support systems. With over three years of experience in web application development, UI/UX optimization, and database-driven solutions, Girija has successfully developed and deployed large-scale web platforms across various domains, including forest management, public health, environmental sustainability, and geospatial decision-making. She has played a crucial role in building dynamic, user-centric web applications that integrate real-time data analytics and geospatial visualization. Her experience extends to developing GIS-enabled dashboards, interactive mapping interfaces, and cloud-based application architectures, which align with this project's focus on forest monitoring, remote sensing, and decision-support tools. Girija's technical expertise spans multiple front-end and back-end technologies, including ASP.NET, ReactJS, JavaScript, jQuery, PostgreSQL, SQL Server, and cloud-hosted environments. She has extensive experience in geospatial web development, working with OpenLayers, Leaflet.js, GeoServer, and PostGIS, which are essential for building interactive GIS-based applications. Her proficiency in database management and API integration allows her to design efficient data pipelines for real-time information retrieval and visualization. She has

	implemented security best practices, including authentication protocols (OAuth 2.0), role-based access controls, and encryption techniques, ensuring that the applications she develops adhere to industry security standards. Additionally, her expertise in responsive web design and accessibility standards ensures that the GIS dashboards and geospatial data platforms are intuitive and user-friendly across multiple devices. She has been instrumental in developing WebGIS solutions for land-use monitoring, deforestation tracking, and afforestation planning, ensuring that policymakers and conservationists can make data-driven decisions based on real-time insights. Her work in integrating Sentinel Hub API services into WebGIS platforms enables seamless satellite imagery processing, vegetation index analysis, and land-cover classification. Her ability to optimize geospatial applications for real-time user interaction, data accuracy, and system responsiveness ensures that the web platform for forest monitoring will operate efficiently under large data loads and complex user interactions. Her role will be critical in ensuring that the web application provides seamless, user-friendly access to forest monitoring data, enabling decision-makers, field officers, and policymakers to effectively track afforestation efforts, patrol routes, and forest cover changes. With her expertise in scalable web development, GIS integration, and user-centric design, she will contribute to delivering a high-performance, secure, and interactive platform for forest conservation in Gujarat.
Mr. Nirmal K Shaji 	Mr. Nirmal K Shaji is a dedicated mobile application developer with over 4 years of experience in building high-performance, user-friendly mobile applications for Android and iOS platforms. He specializes in Flutter, Java, Kotlin, Ionic, and React Native, ensuring cross-platform compatibility, seamless UI/UX, and high-efficiency app performance. He has contributed to Engagement of an IT Agency for ICT Material Development (Agriculture), and Agri Entrepreneur Growth Foundation (AEGF), developing mobile solutions that integrate real-time data processing, offline functionality, and intuitive user interfaces. His expertise in mobile API integration, push notifications, and financial transaction security has been instrumental in enhancing user engagement and mobile accessibility. Having worked on mobile-based solutions in India, Africa, and the USA, he understands the nuances of developing scalable and secure applications that cater to diverse industry

	requirements. His technical skills in mobile architecture, real-time synchronization, and fintech integration make him a critical contributor to cutting-edge mobile technology solutions.
Mr. Mubasheer Ul Islam	Mr. Mubasher Ul Islam is a versatile mobile application developer with over 4 years of experience in Android and iOS development, specializing in mobile UI/UX optimization, high-performance app architecture, and data security. He has expertise in Java, Flutter, HTML, CSS, JavaScript, Python, and PHP, with a strong foundation in database integration using Room Database, SQL Server, and MongoDB. His contributions to Engagement of an IT Agency for ICT Material Development (Agriculture), and TIGRAM – Timber Management and Traceability, Van IT, Millet Market, Bharati Makka, and other application have been vital in delivering efficient, scalable, and secure mobile solutions. His ability to integrate GIS-based tracking, blockchain authentication, and fintech security into mobile applications ensures enhanced usability and seamless performance. With experience spanning India, Africa, and the USA, he has developed cross-industry technology solutions tailored to various mobile user experiences. His expertise in app security, performance optimization, and cloud-based mobile solutions makes him a key player in next-generation mobile application development.
Mr. Mangesh Subhash Nuel	Mangesh Subhash Neul is a highly skilled Quality Assurance (QA) Engineer with a Bachelor's degree in Electronics and Communication Engineering from Kavikulguru Institute of Technology and Science, Ramtek, Nagpur, Maharashtra, India. His strong academic foundation in computer systems, software development life cycle (SDLC), and automated testing frameworks has equipped him with an analytical mindset and deep technical expertise in ensuring software reliability, performance, and security. Mangesh has over three years of hands-on experience in software quality assurance, test automation, and mobile/web application testing. His professional journey has been focused on ensuring system reliability, performance efficiency, and security compliance for large-scale software applications. He has worked on multiple projects related to forestry, healthcare, environmental monitoring, and GIS-based solutions, performing comprehensive functional, regression, and performance testing to ensure high-quality, bug-free application deployments. His role has involved collaborating with development teams to identify vulnerabilities, conducting end-to-end test case execution, and implementing automated test

	scripts to enhance software robustness. His experience in testing GIS-based applications and WebGIS dashboards aligns perfectly with this project's focus on forest monitoring, geospatial data integration, and mobile-based field data collection. As a QA Engineer, Mangesh has developed a strong technical skill set in manual and automated testing, test planning, and system validation. He has expertise in functional testing, black box testing, boundary value analysis (BVA), and equivalence partitioning (ECP) to identify software defects efficiently. His proficiency in Selenium WebDriver, TestNG, and Jenkins enables him to implement automation test suites for continuous integration and deployment (CI/CD) workflows. He is experienced in mobile application testing, ensuring compatibility, performance, and usability of apps in various environments. His experience in performance and load testing helps optimize server response times and scalability, ensuring the WebGIS platform remains efficient and responsive. He has conducted extensive testing of geospatial decision-support tools, verifying the accuracy and reliability of GIS data visualization, map overlays, and spatial analytics. His experience in testing Sentinel Hub API integrations and real-time geospatial data processing ensures the platform's accuracy and responsiveness in forest monitoring applications. His bug-tracking methodologies, defect documentation, and test automation strategies contribute significantly to software stability and performance. He will conduct manual and automated testing for mobile data collection applications, WebGIS-based forest cover change detection systems, and decision-support tools for policymakers. His expertise in automated regression testing will help in identifying system vulnerabilities early, ensuring that the forest management platform operates with high efficiency and minimal downtime. His contributions will be instrumental in delivering a robust, error-free, and high-performance system for real-time forest monitoring and conservation efforts in Gujarat.
Ms. Fathima Begum	Fathima Begum is a highly skilled Deployment Engineer with a Bachelor's degree in Computer Science Engineering from Vignana Bharathi Institute of Technology, JNTU Hyderabad, India. With a strong academic foundation in computer systems, cloud computing, and network security, she has developed deep expertise in deploying, managing, and optimizing cloud-based infrastructure solutions. Her technical education has provided her with a robust understanding of database management, IT security, automation, and scalable application

hosting, all of which are crucial in modern cloud computing environments.

With over three years of professional experience, Fathima has been instrumental in managing IT infrastructure, deploying cloud applications, and optimizing server performance for various large-scale projects. She has played a key role in configuring, deploying, and maintaining cloud-based applications in mission-critical environments, ensuring high availability, security, and scalability. Her work spans multiple domains, including forestry, health monitoring systems, GIS-based environmental platforms, and government-backed digital transformation initiatives. She has successfully led server migration projects, optimized system performance through cloud automation, and implemented high-security access controls for various organizations. Her experience extends to working with geospatial applications, where she has managed the deployment of GIS-based forest monitoring tools, real-time environmental data processing systems, and secure remote sensing applications.

Fathima possesses a diverse set of technical expertise and specialized skills in cloud infrastructure management, IT security, automation, and DevOps workflows. She is well-versed in AWS Cloud Services, including EC2, RDS, S3, IAM, CloudFront, ELB (Elastic Load Balancing), and CloudFormation, making her adept at deploying and maintaining scalable cloud environments. She is proficient in server management and optimization, ensuring high system reliability and efficiency by actively monitoring Ubuntu and Windows servers. Her experience in automated deployment and DevOps tools includes working with CI/CD pipelines, Jenkins, and infrastructure as code (IaC) using CloudFormation and Terraform. She has extensive experience in data backup and disaster recovery planning, including AWS S3 backup management, multi-region RDS instance replication, and high-availability cloud architecture setup.

Throughout her career, Fathima has worked on several large-scale projects, including the deployment of geospatial applications for environmental conservation, GIS-based decision support systems for forestry, and digital platforms for real-time monitoring of natural resources. She has been actively involved in configuring and deploying WebGIS platforms, ensuring seamless integration with Sentinel Hub APIs for real-time satellite data processing. Her work in cloud-based GIS hosting has facilitated the visualization and analysis of environmental data, contributing to forest management,

	afforestation planning, and land-use monitoring. By leveraging her expertise in cloud security, role-based access controls, and encrypted data transmission, she has ensured that these platforms comply with data privacy regulations and cybersecurity best practices. She will be responsible for optimizing AWS cloud infrastructure to support real-time geospatial data processing, integrating Sentinel-2 satellite imagery for forest monitoring, and ensuring the system's high availability and security.
Mr. Siddhartha Goli 	Mr. Siddhartha Goli is a meticulous documentation specialist with expertise in technical writing, compliance documentation, and process standardization. He has been responsible for preparing user manuals, administrative guides, inception reports, and training documentation for complex software systems and technology-driven projects. His work on TIGRAM – Timber Traceability, Agri Entrepreneur Growth Foundation (AEGF), and Van IT has been instrumental in ensuring well-structured, high-quality documentation that enhances knowledge transfer and project transparency. His strong proficiency in Microsoft Office, Advanced Excel, and Adobe Photoshop enables him to create detailed, visually engaging, and easy-to-understand documentation. Having worked in India, he has provided documentation support for various technology-driven initiatives, ensuring that project deliverables are well-documented, standardized, and easily comprehensible for all stakeholders. His role in training, compliance, and knowledge management significantly enhances project efficiency and usability.

1. Technical-Methodological Design

1.1 Strategy

1.1.1 Interpretation of the objectives in the TORs, critical examination of tasks

The RECAP4NDC Project outlined in the Terms of Reference (TOR) aims to develop an integrated Monitoring, Evaluation, and Reporting (MER) system for Forest Landscape Restoration (FLR) initiatives in selected landscapes across India. The central objective is to design and pilot scalable, participatory, and evidence-based MER frameworks, along with digital tools, that can support forest restoration planning and reporting at state and national levels. This will be accomplished through co-development of standardized MER indicators, methods, and tools; application of modern technologies such as remote sensing, GIS, drones, IoT, and AI; and stakeholder-driven institutional integration.

A key focus of the project is on creating MER systems that are aligned with India's national priorities and international commitments, such as the Bonn Challenge, UN SDGs, Paris Agreement, State of Forest Reports, and National Forest Inventory. By designing standardized frameworks and protocols, the project seeks to enable consistent, comparable, and outcome-focused monitoring across diverse restoration contexts. Additionally, the assignment requires integration with existing national and sub-national forest governance systems to ensure uptake and sustainability.

Another significant goal is the development and deployment of digital tools including mobile applications, dashboards, and GIS platforms to collect, visualize, and analyze restoration data. These tools should support real-time data collection, remote validation, and interactive reporting for various stakeholders, from field-level practitioners to state and central policymakers. The digital solutions must be user-friendly, flexible for use in different landscapes, and interoperable with existing systems, enabling seamless data exchange and institutional adoption.

Furthermore, the assignment emphasizes piloting the MER framework and tools in selected landscapes across Gujarat, Maharashtra, Uttarakhand, and Delhi NCR. Field testing and refinement will be critical to ensure that the systems are functional, scalable, and context-appropriate. The pilots will demonstrate how MER frameworks and digital tools can track ecological outcomes (e.g., forest cover, soil health), social outcomes (e.g., livelihoods, participation), and institutional outcomes (e.g., policy alignment, stakeholder coordination) of restoration interventions.

The assignment also underscores the need for participatory development of all outputs. This includes engaging key national and sub-national stakeholders such as MoEFCC, FSI, SFDs, IUCN, GIZ, and restoration practitioners through interviews, consultations, and workshops. This participatory approach ensures contextual relevance, capacity building, and stakeholder ownership of the MER systems and tools.

In addition, the project calls for strong knowledge management and capacity-building components. This includes developing user manuals, SOPs, and training modules for field staff and government officials. Capacity-building efforts such as training-of-trainers programs and regional workshops will be necessary to build long-term institutional capacity to use and manage the MER systems.

The assignment has the following specific objectives:

- Design of integrated and standardized Monitoring, Evaluation, and Reporting (MER) frameworks for Forest Landscape Restoration (FLR).
- Co-development of restoration indicators, methods, and tools aligned with national and global goals.
- Design and deployment of digital tools such as mobile apps, dashboards, and GIS platforms for MER.
- Field testing and piloting of the MER framework and tools in selected FLR landscapes.
- Stakeholder engagement and participatory development of MER systems.
- Capacity building and knowledge management for sustainable institutional adoption.
- Integration of Sentinel Hub APIs for High-Resolution Satellite Imagery Processing.

1.1.2 Contextualization for Gujarat State

The proposed solution is specifically designed to meet the needs of the Gujarat State Rural Livelihood Mission (GSRLM) under the Government of Gujarat. The modules including Grievance Redressal, Geo-tagging, and Knowledge Hub are tailored to support Self-Help Groups (SHGs), Village Organizations (VOs), and Cluster-Level Federations (CLFs) operating in the state's diverse rural landscape. Special attention will be given to incorporating Gujarati language support, integrating with existing Gujarat-specific MIS systems, and ensuring usability for rural women and functionaries with varying levels of digital literacy.

1.1.3 Description and Justification of the contractors strategy for delivering the services put out of tender

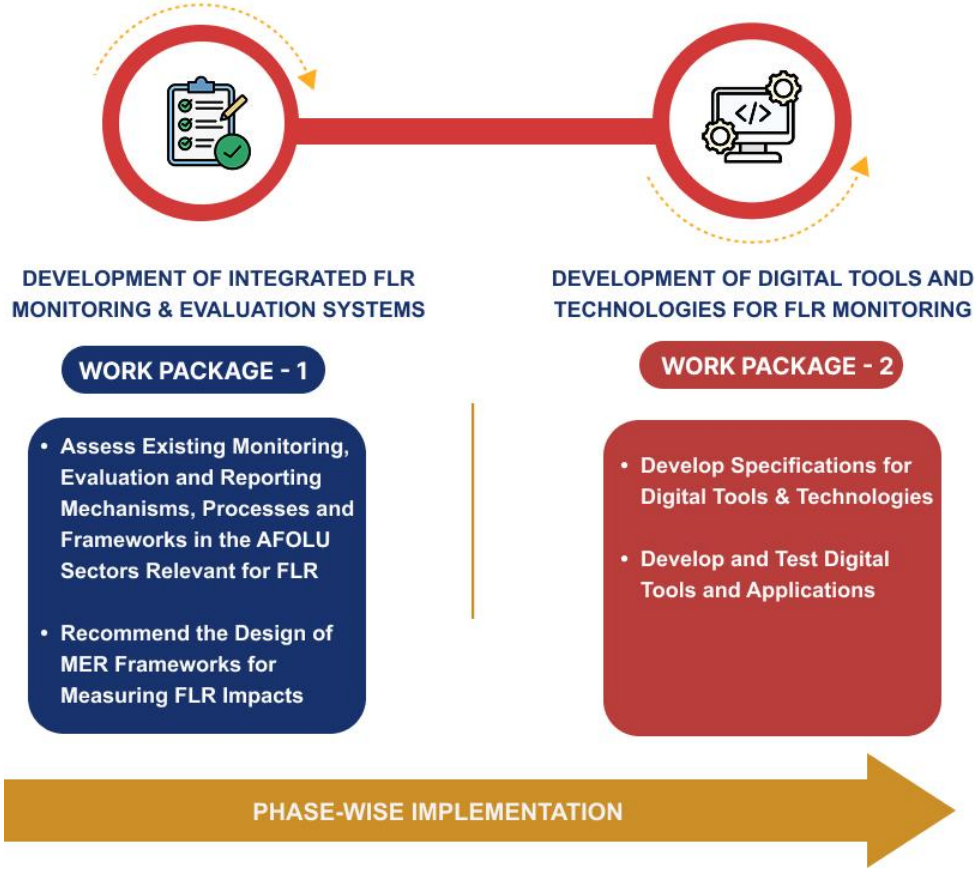
The consortium led by GISFY Private Limited, with Everain Global Services Private Limited as a strategic partner, proposes a rigorous, participatory, and technology-enabled approach to deliver the services under the RECAP4NDC assignment. The strategy is anchored in the consortium’s deep experience in ecological monitoring, geospatial system development, digital MER tools, and institutional engagement.

The overall implementation strategy is designed to support the development of a robust Monitoring, Evaluation, and Reporting (MER) system for Forest Landscape Restoration (FLR) in India. This includes both the technical development of frameworks and digital tools, and the human and institutional capacity-building required to ensure long-term adoption and sustainability.

We structure our delivery around the two Work Packages specified in the TOR:

- Work Package 1: Development of an integrated FLR monitoring and evaluation system in India.
- Work Package 2: Development of digital tools and technologies to operationalize the MER system and its institutional adoption.

Each Work Package consists of targeted activities that are mutually reinforcing and responsive to the TOR's requirements.



Work Package 1: Development of Integrated FLR Monitoring & Evaluation Systems

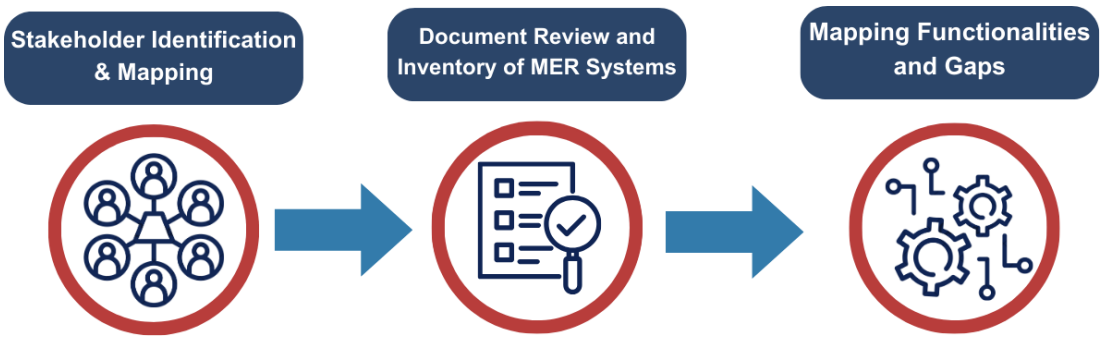
This work package aims to create a robust and integrated Monitoring, Evaluation, and Reporting (MER) system for Forest Landscape Restoration (FLR) across India. It focuses on building upon existing MER frameworks within the Agriculture, Forestry, and Other Land Use (AFOLU) sectors and tailoring them to monitor FLR outcomes. This work will support national commitments under the Bonn Challenge, Land Degradation Neutrality targets and NDC goals.

Activity 1.1: Assess Existing Monitoring, Evaluation and Reporting Mechanisms, Processes and Frameworks in the AFOLU Sectors Relevant for FLR

This activity will systematically analyze current MER systems in place across AFOLU sectors to evaluate their alignment with FLR indicators and outcomes. The goal is to identify strengths, weaknesses, and opportunities for harmonization to ensure a cohesive FLR monitoring structure across India.

Task 1.1.1: Stakeholder Mapping and Inventory of Existing MER Frameworks

This task involves a comprehensive review of stakeholders and existing MER frameworks.



Stakeholder Identification and Mapping

- Identify key stakeholders from national, state, and district levels across AFOLU sectors. This includes ministries (MoEFCC, MoA&FW), technical agencies (FSI, NRSC), and NGOs involved in restoration.
- Map institutional mandates and roles related to data collection, monitoring, and reporting, focusing on overlaps and gaps relevant to FLR.
- Engage stakeholders through interviews, focus group discussions, and participatory workshops to document their current and potential roles in FLR monitoring.
- Evaluate communication channels and coordination structures between institutions involved in MER.
- Utilize GIS to visualize spatial presence and operational zones of stakeholders.
- Document the historical engagement of each stakeholder in MER systems and FLR initiatives.
- Analyze capacity levels, funding, and technical expertise available with stakeholders.
- Categorize stakeholders by influence, interest, and institutional readiness for collaboration in the proposed FLR monitoring system.

Document Review and Inventory of MER Systems

- Collect policy guidelines, manuals, annual reports, technical standards, and existing toolkits related to MER in forestry, agriculture, and watershed programs.
- Analyze spatial and temporal coverage of monitoring systems, including integration with geospatial platforms such as Bhuvan, VEDAS, and SEPAL.
- Examine indicators, data standards, reporting intervals, and compliance mechanisms adopted under each MER framework.
- Assess the presence of monitoring technologies like mobile apps, drones, and satellite imagery within these systems.
- Identify reporting requirements for national and international commitments (e.g., SDGs, UNCCD, UNFCCC, CBD).
- Create a repository of datasets, including metadata such as spatial resolution, periodicity, and responsible agency.
- Document interoperability challenges among existing digital MER platforms.
- Highlight best practices and case studies that can inform the design of an FLR-centric MER framework.

Mapping Functionalities and Gaps

- Develop a comparative matrix of key functionalities across MER systems, evaluating efficiency, inclusiveness, and scalability.
- Identify data collection duplication, inconsistent methodologies, and fragmented reporting lines that hinder effective FLR monitoring.
- Examine institutional silos that impede coordination and data sharing across ministries and departments.
- Analyze technological and infrastructure barriers to real-time monitoring, particularly in remote and tribal areas.
- Assess user-friendliness and field applicability of existing MER tools, identifying needs

for capacity building.

- Review institutional incentives and mandates that may impact data accuracy or compliance with MER procedures.
- Highlight areas where frameworks meet international standards and where localization is needed.
- Recommend areas for consolidation, integration, or reengineering of current systems to improve alignment with FLR objectives.

Task 1.1.2: Analytical Review of MER Frameworks and Institutional Capacities

This task involves a detailed analysis of institutional capacities and performance indicators.

Institutional Capacity Assessment

- Conduct a detailed analysis of institutional roles, mandates, and capacities for MER functions relevant to FLR.
- Evaluate the availability of technical personnel, digital infrastructure, and logistics at national and state levels.
- Identify capacity gaps in terms of knowledge, skill sets, and resource allocation among implementing agencies.
- Review previous MER training programs and assess their applicability to FLR-specific needs.
- Analyze coordination structures and governance challenges affecting institutional performance in MER.
- Examine budget allocations and funding mechanisms supporting MER activities across departments.
- Determine the potential for digital transformation and automation in MER functions.
- Propose a capacity development roadmap for scaling FLR monitoring through institutional strengthening.

Performance Indicator Analysis

- Compile and categorize indicators currently used in forestry, watershed, and agriculture programs.
- Evaluate indicator sensitivity to restoration outcomes like biomass increase, biodiversity enhancement, and land degradation neutrality.
- Assess consistency in definitions, baseline measurements, and periodicity across departments.
- Determine the extent of disaggregated data availability (gender, geography, livelihoods).

- Examine whether indicators are outcome-focused or activity-centric, and their alignment with global standards.
- Identify indicators that can serve as proxies for ecosystem services and resilience.
- Recommend new indicators for FLR integration based on scientific validity and field practicality.
- Propose standardized templates for indicator tracking, reporting, and impact assessment.

Global Benchmarking

- Review internationally recognized MER systems, including CIFOR's FLR monitoring guidelines and FAO's SEPAL platform.
- Identify technological, operational, and institutional features that have proven successful in other regions.
- Evaluate adaptability of these systems to Indian federal structures, data needs, and program cycles.
- Compare open-source versus proprietary platforms for scalability and cost-effectiveness.
- Investigate policy and incentive frameworks that support MER adoption in other countries.
- Draw lessons from community monitoring initiatives and citizen science models globally.
- Develop a benchmark matrix comparing Indian systems with global counterparts.
- Formulate recommendations for integrating best practices into India's proposed FLR MER system.

Task 1.1.3: Synthesis and Development of a Gap Analysis Report

This task involves synthesizing findings and developing a gap analysis.

Draft Gap Analysis Report

- Synthesize insights from the Stakeholder Mapping, Inventory of existing MER frameworks, and the analytical review of MER frameworks and institutional structures into a comprehensive narrative that outlines the current status and effectiveness of existing MER frameworks.
- Identify key gaps in data collection, institutional mandates, technological infrastructure, and interdepartmental coordination.
- Provide a thematic breakdown of gaps by sector forestry, agriculture, land, and water and by geographic zone.

- Structure the report with visual elements like graphs, matrices, and infographics to enhance understanding.
- Outline strategic areas for harmonization with national environmental and climate-related reporting platforms.
- Integrate case studies from previous successful restoration monitoring initiatives in India and abroad.
- Ensure that the report addresses both operational and strategic dimensions of the MER ecosystem.
- Position the report as a foundational document to guide the design of the FLR-specific MER framework in subsequent activities.

Stakeholder Validation

- Design a participatory validation process engaging representatives from key ministries, line departments, and civil society.
- Organize regional consultations and national workshops to review and validate the findings of the gap analysis report.
- Use interactive methods such as SWOT analysis, problem tree analysis, and participatory ranking tools.
- Capture stakeholder inputs through surveys, group discussions, and plenary feedback sessions.
- Address diverse perspectives, especially from local implementation agencies and community-based organizations.
- Document all feedback with attribution to inform a transparent revision process.
- Revise the draft gap analysis report based on inputs received and prepare a final validated version.
- Ensure that the validated report is accepted and endorsed by key nodal agencies for official integration into planning.

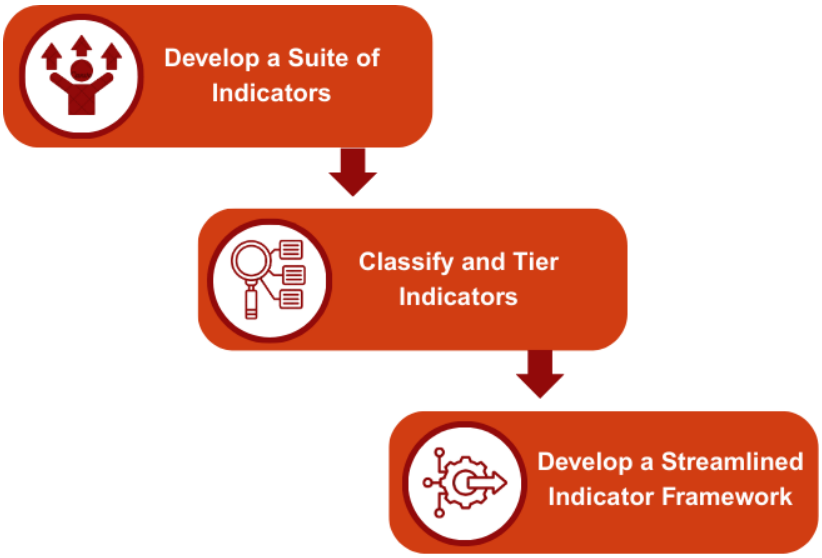
Final Synthesis and Roadmap

- Develop a strategic roadmap that outlines short-term, medium-term, and long-term interventions for bridging identified gaps.
- Assign responsibilities to institutional actors for each phase of implementation based on their mandates and capacities.
- Propose timelines, budgets, and indicators for monitoring progress on roadmap implementation.
- Integrate capacity-building and technology adoption strategies into the roadmap.
- Align the roadmap with existing policy frameworks such as the National Action Plan on Climate Change (NAPCC), NDCs, and SDG 15.3 targets.
- Highlight potential areas for donor and multilateral engagement to support implementation.
- Include contingency planning for operational risks, funding limitations, or stakeholder resistance.
- Present the roadmap in a modular format to allow phased implementation and customization by state governments.

Activity 1.2: Recommend the Design of MER Frameworks for Measuring FLR Impacts

This activity focuses on designing new and improved MER frameworks tailored for FLR that build upon the findings from Activity 1.1. It will propose specific indicators, data workflows, governance mechanisms, and reporting formats necessary to measure the ecological, social, and economic outcomes of FLR efforts in India.

Task 1.2.1: Define FLR-specific Indicators and Metrics



Develop a Suite of Indicators

- Identify biophysical indicators such as forest cover, biomass, soil carbon, land productivity, and water availability to measure ecosystem restoration.
- Include socio-economic indicators such as household income diversification, employment in restoration activities, and gender inclusion metrics.
- Consider institutional indicators like governance effectiveness, inter-agency collaboration, and policy alignment.
- Ensure indicators are scalable and usable at national, state, and landscape levels.
- Validate indicator selection with stakeholders to ensure relevance and measurability.
- Align indicators with SDGs (particularly 15.3), Bonn Challenge Barometer, and India's NDC framework.
- Propose core vs. optional indicators based on implementation contexts.
- Ensure indicators support adaptive management and decision-making.

Classify and Tier Indicators

- Establish a tiered structure with national-level, state-level & landscape-level indicators.
- Specify data collection methods (field measurements, remote sensing, etc.) for each tier.
- Outline how local data will be aggregated to inform national reporting.
- Define spatial and temporal scales for measurement and reporting.
- Clarify roles of different stakeholders in data collection and aggregation.
- Allow flexibility for states to include additional indicators relevant to their context.

Develop a Streamlined Indicator Framework

- Consolidate a comprehensive list of potential indicators into a core set for regular monitoring.
- Prioritize indicators that are cost-effective, easy to measure, and policy-relevant.
- Ensure the framework aligns with national and international reporting requirements.
- Incorporate feedback loops for periodic review and revision of the indicator set.
- Document the scientific basis and rationale for selecting each indicator.

Task 1.2.2: Design Data Collection Methodologies and Protocols

This task includes the following sub-tasks:

1. Develop Standardized Protocols
   - Develop detailed, step-by-step protocols for data collection, including sampling design, measurement techniques, and frequency.
   - Incorporate quality assurance and quality control (QA/QC) mechanisms to ensure data accuracy and reliability.
   - Outline standard operating procedures (SOPs) for data entry, storage & management.
   - Ensure protocols are adaptable to diverse field conditions and resource availability.

2. Select Appropriate Data Collection Technologies
   - Evaluate the use of remote sensing, drones, mobile apps, IoT devices, and other technologies for data collection.
   - Assess the feasibility and cost-effectiveness of different technologies in the Indian context.
   - Provide guidance on technology selection based on the specific indicator, scale of measurement, and available expertise.

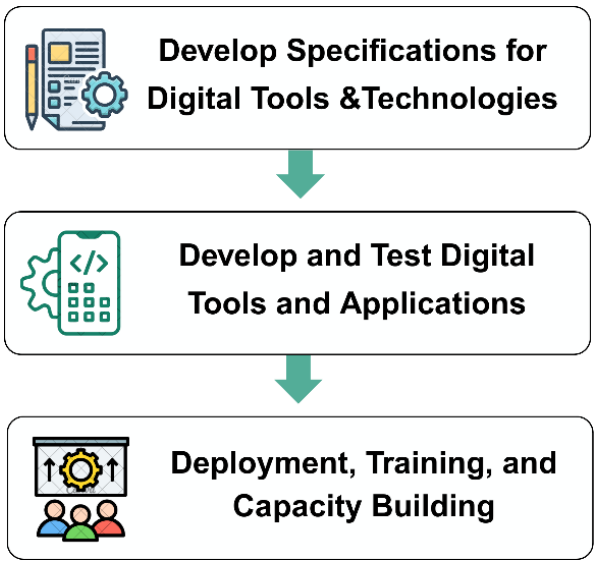
3. Design a Scalable Data Management System
   - Develop a centralized database architecture for storing, managing, and analyzing FLR data.
   - Ensure data security, integrity, and accessibility for authorized users.
   - Establish data sharing agreements and protocols among different stakeholders.
   - Enable interoperability with existing national and state-level data platforms.

Task 1.2.3: Outline Institutional Arrangements and Governance Mechanisms

1. Define Roles and Responsibilities
   - Clearly define the roles and responsibilities of different government agencies, research institutions, NGOs, and local communities in the MER process.
   - Establish mechanisms for inter-agency coordination and collaboration.
   - Define data sharing agreements and protocols to facilitate information exchange.
2. Establish Governance Structures
   - Propose a multi-stakeholder governance framework for overseeing the implementation of the FLR MER system.
   - Outline decision-making processes, including mechanisms for stakeholder input and conflict resolution.
   - Ensure accountability and transparency in data collection, analysis, and reporting.
3. Develop Feedback and Adaptive Management Mechanisms
   - Design mechanisms for data feedback and dissemination to inform policy and practice.
   - Incorporate adaptive management principles into the MER system to allow for adjustments based on new information and changing conditions.
   - Ensure that the MER system supports continuous improvement in FLR implementation.

Work Package 2: Development of Digital Tools and Technologies for FLR Monitoring

This work package focuses on developing the digital tools necessary to support the implementation of the FLR monitoring framework designed in Work Package 1. This includes creating software applications, data dashboards, and decision support systems to facilitate data collection, analysis, visualization, and reporting.



Activity 2.1: Develop Specifications for Digital Tools and Technologies

This activity involves defining the technical requirements, functionalities, and user needs for the digital tools to be developed.

Technical Sections for the Knowledge Exchange Portal (KExP)

Work Package 1: Design, Develop, and Deploy a Fully Functional, Bilingual (English and Hindi) Web Portal for FLR Knowledge Management and Stakeholder Engagement

This work package outlines the comprehensive approach to designing, developing, and deploying a robust, user-centric, and secure Knowledge Exchange Portal (KExP). Our methodology ensures a highly functional, bilingual platform that serves as a central hub for FLR knowledge, fostering collaboration and capacity building among diverse stakeholders.

Activity 1.7: Provide easy navigation and search functionalities for users to access information by theme or region. Integrate functionalities for hosting or linking webinars, knowledge-sharing events, and recordings. Create user categories-based access permissions for example, based on stakeholder types.

To achieve intuitive navigation, powerful search, dynamic event management, and granular access control, we propose the following:

* **Information Architecture and Navigation:** We will define a clear and logical information architecture (IA) through sitemaps and user flow diagrams. Content will be intuitively grouped into overarching themes (e.g., Agroforestry, Community Engagement, Policy, Financing) and geographic lenses (e.g., National, State, District, Landscape). Breadcrumb trails will be implemented on all interior pages to provide users with a clear sense of their location within the portal.
* **Bilingual Taxonomy and Metadata Schema:** A comprehensive, bilingual (English and Hindi) taxonomy and metadata schema will be established. This schema will include mandatory tags for theme, subtheme, geography (country/state/district/landscape), ecosystem type, target audience, and resource type. These tags will be enforced within the Content Management System (CMS) at the time of content publication to ensure consistency and discoverability.
* **Responsive Global Navigation:** A highly responsive global navigation system will be designed, featuring a mega-menu for desktop users that exposes top-level themes, regions, and frequently used tools. On interior pages, a secondary sidebar navigation will provide contextual options. A persistent, easily accessible search bar with quick suggestions will be integrated across the portal for immediate access to information.
* **Faceted Search with Elasticsearch:** We will implement a powerful faceted search capability leveraging Elasticsearch. This will allow users to refine their search results using multiple filters such as keyword, theme, region, resource type, year, and author. Multi-select filters, real-time result counts, and interactive "chips" for easy filter removal will enhance the user experience.
* **Geo-aware Discovery:** To facilitate region-specific knowledge discovery, we will configure geo-aware functionalities. This includes interactive region pickers, clickable India map overlays (potentially integrating with GIS data from Activity 1.5), and the use of URL query parameters (e.g., `?state=Gujarat&theme=Agroforestry`) to enable shareable, deep-linked search results.
* **Optimized Search Relevance:** Search relevance will be optimized through the use of language-specific analyzers for both English and Hindi (including stemming, stopwords, and synonyms). Recent and curated items will be boosted in search results, and spell-tolerant matching will be implemented. Search queries will be logged to continuously refine synonyms and improve search accuracy over time.
* **Saved Searches and Alerts:** Users will have the ability to save their search queries and subscribe to alerts (via email or in-app notifications) for specific themes or regions. This ensures they are automatically notified of new resources or events relevant to their interests.
* **Accessibility in Navigation and Search:** We will ensure full compliance with WCAG 2.1 AA guidelines for navigation and search. This includes keyboard-navigable menus, appropriate ARIA roles, skip-links for content, descriptive labels, and high-contrast focus states for interactive elements.
* **Contextual Navigation:** Content pages will feature contextual navigation elements such as "related items" based on shared taxonomy, "more from this region/theme," and breadcrumbs that dynamically reflect the active filter path, enhancing content discoverability.

* **Unified Events Hub:** A comprehensive Events hub will be integrated, offering both calendar and list views. Users will be able to filter events by theme, region, hosting partner, and format (webinar/workshop). iCal feeds will be exposed for easy calendar synchronization, and all timestamps will be time-zone aware.
* **Webinar Hosting/Embedding:** The portal will support webinar hosting or embedding via approved third-party providers (e.g., Zoom, Microsoft Teams, YouTube Live). This includes secure join links, registration functionalities, waiting lists, and automated reminders. Post-event recordings will be securely stored with rich metadata.
* **Automated Event Asset Generation:** The system will automatically generate essential event assets, including dedicated registration pages, speaker bios, downloadable agendas, and "Add to Calendar" (ICS) buttons. Integrated forms will capture attendance and feedback for post-event analysis.
* **Rich Recording Publication:** Published recordings will include transcripts, captions in both English and Hindi, chapter markers for easy navigation, and downloadable slide decks. Each recording will be linked to its original event page and related resources based on theme and region.
* **Event Content Lifecycle:** A defined content lifecycle for events (draft ? scheduled ? live ? recorded) will be implemented, incorporating moderation checkpoints at each stage. "Upcoming" and "past events" will be prominently surfaced on relevant theme and region landing pages.
* **Role-Based Access Control (RBAC):** We will design a robust RBAC system with predefined stakeholder categories (e.g., public guest, registered user, practitioner, researcher, government/SFD, editor, moderator, admin). Each role will be mapped to least-privilege permissions, ensuring users only access what is necessary.
* **Permissions Matrix:** A detailed permissions matrix will be defined, governing actions such as viewing/downloading restricted resources, posting in forums, submitting events, accessing draft translations, and administering taxonomy. These permissions will be enforced via middleware on every request to the backend.
* **Fine-Grained Content Visibility:** The portal will support fine-grained content visibility using taxonomy-linked policies. For example, resources tagged "internal?Govt" will only be visible to users with government/SFD roles. Support for embargo dates and expiring access will also be included.
* **Self-Service Role Requests and Audit Logs:** Users will be able to submit self-service role requests, which will undergo an approval process (with justification and verification for government/academic emails). Comprehensive audit logs will be maintained for all role changes and content access events, ensuring accountability.

Activity 1.8: Integrate accessibility features in compliance with Web Content Accessibility Guidelines 2.1 (WCAG 2.1) guidelines.

Accessibility will be a foundational principle throughout the KExP's development lifecycle, ensuring an inclusive experience for all users.

* **Accessibility Requirements Analysis:** At the project inception, we will conduct a thorough accessibility requirements analysis. This will involve mapping WCAG 2.1 Level AA success criteria directly to all portal features, including navigation, search, resource library, GIS maps, and e-learning modules, to ensure comprehensive coverage.
* **Semantic HTML5 Structure:** We will adopt a clean and semantic HTML5 structure, utilizing proper landmark roles, heading hierarchies, and ARIA roles judiciously and only where necessary to enhance meaning without introducing redundancy.
* **Keyboard Operability:** All interactive components (buttons, menus, accordions, modals, carousels) will be fully operable via keyboard navigation. This includes ensuring a logical tab order, highly visible focus indicators, and "skip-to-content" links for efficient navigation.
* **Sufficient Color Contrast:** We will ensure sufficient color contrast across the entire portal, testing palettes with automated tools to meet WCAG 2.1 AA standards. We will avoid relying solely on color to convey meaning, providing alternative visual cues.
* **Scalable Typography:** Scalable typography and relative units (em/rem/%) will be used throughout the design to ensure text reflows gracefully without loss of content or functionality, even when zoomed up to

200%.
* **Screen Reader Compatibility:** The portal will be designed for full screen reader compatibility, incorporating descriptive labels, comprehensive alt text for all images, appropriate ARIA attributes for dynamic content updates, and live-region alerts for notifications.
* **Bilingual Content with Lang Attributes:** Bilingual content (English/Hindi) will be implemented with proper `lang` attributes, enabling screen readers to accurately pronounce text and render scripts correctly.
* **Captions, Transcripts, and Sign Language:** All video recordings will include accurate captions. Transcripts will be provided for all audio content. Where feasible and deemed critical for key training modules or high-level events, sign language interpretation will be considered.
* **Accessible GIS Map Interfaces:** GIS map interfaces will be validated for accessibility, providing keyboard-operable controls, filtering options, and tabular data downloads as accessible alternatives for users unable to interpret maps visually.
* **Accessible Forms:** All forms (registration, feedback, quizzes) will include clear, associated labels, descriptive error messages, explicit instructions, and proper focus management for validation errors, guiding users effectively.
* **Avoidance of Time Limits:** We will avoid imposing time limits wherever possible. If time limits are absolutely necessary, users will be provided with clear warnings, options to extend the time, or the ability to disable timeouts.
* **Iterative Accessibility Testing:** Accessibility will be tested iteratively throughout the development process using automated tools (axe, Lighthouse, WAVE) and manual audits with leading screen readers (NVDA, JAWS, VoiceOver) across various devices and browsers.
* **User Engagement in Usability Testing:** We will actively engage users with disabilities in usability testing sessions to validate the real-world accessibility of critical workflows such as navigation, search, forums, and e-learning modules.
* **Accessibility Conformance Report (ACR):** A dedicated Accessibility Conformance Report (ACR) will be documented, following the WCAG template, and updated with each major release cycle to demonstrate ongoing compliance.

Activity 1.9: Set up a web analytics dashboard for performance and user tracking.

A robust web analytics dashboard will be established to provide deep insights into portal performance and user behavior, supporting data-driven decision-making and continuous improvement.

* **Define Analytics Objectives and KPIs:** We will define clear analytics objectives aligned with project KPIs, such as user engagement, content consumption, event participation, GIS map interactions, and e-learning module completion rates.
* **Analytics Platform Selection:** We will select an appropriate analytics platform, such as Matomo or GA4, based on project requirements, ensuring it supports bilingual content, customizable dashboards, and robust data privacy features.
* **Tracking Script Configuration:** Tracking scripts will be configured within the portal codebase, meticulously ensuring compliance with user privacy, cookie consent mechanisms, and relevant data protection regulations (GDPR/DPDP).
* **Event Tracking:** Comprehensive event tracking will be implemented for all key user interactions, including resource downloads, forum posts, search queries, event registrations, feedback form submissions, and quiz completions.
* **Custom Dimensions and Metrics:** Custom dimensions and metrics will be configured to segment and measure user engagement by stakeholder type, providing granular insights into how different user categories interact with the portal.
* **Thematic and Geographic Metadata Tagging:** Thematic and geographic metadata will be tagged within the content repository. This allows analytics to attribute visits and downloads to specific FLR themes or regions, providing valuable content performance insights.
* **Funnel Analysis:** Funnel analysis will be set up to track critical user journeys, such as "homepage ? knowledge repository ? content download" or "event listing ? registration ? attendance," identifying potential

drop-off points.
* **Performance Monitoring Metrics:** Key performance monitoring metrics will be tracked, including page load time, bounce rates, device/browser distribution, and the geographic distribution of users, to optimize the user experience.
* **CMS Integration for Content Managers:** Analytics will be integrated with the CMS to provide content managers with direct visibility into most viewed/least viewed resources, inactive content, and search terms yielding no results, facilitating content strategy.
* **Role-Based Dashboards:** Dashboards will be designed with role-based views: high-level KPIs for project managers, detailed usage metrics for content managers, and system performance data for developers, ensuring relevant information is accessible to each stakeholder.
* **Automated Reporting:** Automated weekly/monthly reports will be configured for email distribution to GIZ and consortium partners, highlighting trends, anomalies, and areas requiring intervention.
* **Real-time Monitoring Panels:** Real-time monitoring panels (e.g., within Grafana, Matomo, or GA4) will be set up to detect traffic spikes, API errors, or unusual behavior, enabling rapid troubleshooting and incident response.
* **Analytics Setup Validation:** The analytics setup will be rigorously validated with test events, cross-checking logs against dashboard outputs to ensure accuracy and completeness of data.
* **Administrator and Content Manager Training:** Comprehensive training will be provided to administrators and content managers on interpreting analytics dashboards and leveraging insights for continuous content improvement and strategic decision-making.
* **Analytics Governance Documentation:** Analytics governance will be thoroughly documented, including data retention periods, user privacy measures, and standard reporting formats, ensuring long-term sustainability and compliance.

Activity 1.10: Ensure that the portal adheres to international web accessibility as well as cybersecurity standards. Further, the portal must be deployed in a secure hosting environment with robust data protection protocols.

Security and data protection are paramount. We will implement a multi-layered approach to ensure the KExP is resilient against threats and compliant with international standards.

* **Baseline Compliance Assessment:** We will conduct a baseline compliance assessment, mapping all portal requirements against WCAG 2.1 Level AA, ISO 27001, OWASP Top 10, GDPR/DPDP Act, and National e-Governance Standards (NeGS) to identify and address all regulatory and security obligations.
* **Accessibility by Design:** Accessibility will be embedded in every stage of design and development, utilizing semantic HTML, appropriate ARIA attributes, rigorous screen reader testing, keyboard navigation, and automated color-contrast validation tools.
* **Strong Authentication & Authorization:** Robust authentication and authorization mechanisms will be implemented, including mandatory Two-Factor Authentication (2FA) for administrative roles, comprehensive Role-Based Access Control (RBAC), and least-privilege permissions for all administrators and content contributors.
* **Encrypted Communications:** All communications will be encrypted using TLS 1.2+ (with SSL certificates), HSTS headers will be enabled, and secure cookies will be enforced for session management to prevent eavesdropping and session hijacking.
* **Data Encryption at Rest and in Transit:** Sensitive data at rest will be protected using AES-256 encryption. All user credentials will be stored using industry-standard hashing and salting techniques.
* **Environment Separation and Network Isolation:** Separate, isolated environments will be maintained for development, testing, and production. Application, database, and file storage servers will be isolated behind firewalls with integrated intrusion detection and prevention systems (IDS/IPS).
* **Certified Hosting Environment:** The portal will be deployed on ISO 27001 and SOC 2 Type II?certified data centers, ensuring high standards of security, redundancy, and uptime SLAs. Data residency will comply with both Indian and EU laws.
* **Periodic Penetration Testing and Vulnerability Scanning:** Periodic penetration testing (e.g., using

external security firms) and continuous vulnerability scanning (using tools like OWASP ZAP, Burp Suite, Trivy) will be conducted to proactively identify and remediate threats before go-live and throughout the portal's lifecycle.
* **Web Application Firewall (WAF) and DDoS Protection:** A Web Application Firewall (WAF) will be configured to protect against common web exploits. DDoS protection will be implemented, and IP whitelisting will be used for sensitive backend services.
* **Robust Password Policy:** A robust password policy will be enforced, requiring a minimum of 10 characters, complexity requirements (uppercase, lowercase, numbers, special characters), and regular expiration cycles. Accounts will be locked after repeated failed login attempts.
* **Secure Coding Practices (SAST/DAST):** Secure coding practices will be enforced, with automated Static Application Security Testing (SAST) and Dynamic Application Security Testing (DAST) integrated into CI/CD pipelines to identify vulnerabilities early.
* **Comprehensive Audit Trails and System Logs:** A detailed audit trail and comprehensive system logs will be maintained, capturing all logins, data access, updates, and administrative activities for accountability, compliance, and forensic investigation.
* **Backup and Disaster Recovery:** Robust backup and disaster recovery plans will be established, including defined Recovery Point Objective (RPO) and Recovery Time Objective (RTO) targets. Backups will be encrypted and stored across multiple locations, with regular restore testing.
* **Data Protection by Design and Default:** Data protection principles will be embedded by design and by default, conducting privacy impact assessments, minimizing personal data collection, and obtaining explicit user consent for data processing.
* **Security Operations Manual and Incident Response:** Administrators will be provided with a documented security operations manual and an incident response playbook to enable quick and effective action in case of security breaches.
* **Regular Updates and Patch Management:** All system components, libraries, and dependencies will be regularly updated. A comprehensive patch management process will be implemented to promptly address and close known vulnerabilities.
* **Cybersecurity Training:** Administrators and content managers will receive training on cybersecurity best practices, phishing awareness, and secure content handling protocols to foster a security-conscious culture.
* **Compliance Certification and Reporting:** Compliance will be certified through accessibility audits, GDPR readiness checks, and verification of ISO 27001 hosting provider certifications. All relevant reports will be submitted to GIZ for transparency.

Activity 1.11: Conduct usability testing with representative stakeholder groups and incorporate feedback to ensure that the portal is user friendly and optimized for various devices. Final deployment on a secure, reliable hosting platform with secure backup and disaster recovery protocols.

User-friendliness and reliability are critical for the KExP's success. This activity focuses on validating the user experience and ensuring a robust, production-ready deployment.

* **Stakeholder Identification for Usability Testing:** We will identify and recruit representative stakeholder groups for usability testing, including policymakers, State Forest Department officials, researchers, civil society actors, and community practitioners, ensuring diverse perspectives.
* **Structured Test Scenarios and User Journeys:** Structured test scenarios and user journeys will be developed to evaluate critical functionalities. Examples include searching for a specific resource, registering for an event, accessing GIS maps, and completing an e-learning module.
* **Bilingual Prototype Testing Across Devices:** Bilingual prototypes (English/Hindi) will be tested across multiple devices (desktop, tablet, mobile) and various browsers. Testing will also simulate varying bandwidth conditions to ensure optimal performance in real-world usage.
* **Mixed-Method Usability Testing:** A combination of usability testing methods will be employed, including think-aloud protocols, moderated testing sessions, and remote unmoderated testing, to capture both qualitative insights and quantitative user behavior data.
* **Key Usability Metrics Tracking:** Key usability metrics will be tracked, such as task completion rate, time

on task, error rate, navigation clarity, accessibility compliance, and overall user satisfaction scores.
* **Qualitative and Quantitative Data Collection:** Both qualitative and quantitative data will be collected from participants through direct observation, structured feedback forms, and post-test interviews to gain a comprehensive understanding of their experience.
* **Usability Testing Report:** Findings will be synthesized into a comprehensive usability testing report, highlighting identified usability barriers, accessibility issues, and concrete enhancement opportunities.
* **Iterative Design and Development:** Stakeholder feedback will be incorporated into iterative design and development sprints, continuously refining navigation flows, content layouts, and interactive features until a consensus on optimal usability is achieved.
* **Follow-up Validation Testing:** A follow-up round of testing will be conducted to validate the final refinements, confirm the resolution of identified issues, and ensure the portal's readiness for deployment.
* **Secure, ISO 27001-Certified Hosting:** The portal will be deployed on a secure, ISO 27001-certified hosting environment with geographically redundant data centers to ensure high availability and resilience.
* **Secure Backup Systems:** Secure backup systems will be implemented, featuring scheduled incremental and full backups. All backups will be stored in encrypted form across multiple secure locations.
* **Defined RPO/RTO and Failover Testing:** Recovery Point Objective (RPO) and Recovery Time Objective (RTO) targets will be clearly defined for disaster recovery. Failover mechanisms will be regularly tested to ensure business continuity in the event of an outage.
* **Automated Monitoring for Uptime and Performance:** Automated monitoring will be configured for uptime, server performance, and security events, with alerts set up for proactive incident management and rapid response.
* **Hosting Architecture Documentation and Handover:** Comprehensive documentation of the hosting architecture, backup schedules, and disaster recovery protocols will be provided and handed over to administrators for long-term operations.
* **Administrator and Technical Staff Training:** Administrators and technical staff will receive thorough training on recovery procedures, backup verification, and escalation workflows to strengthen institutional resilience and operational readiness.

Work Package 2: Integrate Existing FLR Training Modules, Resources and Embedding External Resources, Online Knowledge Exchange Events onto the Web Portal in a User-Friendly and Easily Searchable Manner

This work package focuses on populating the KExP with valuable FLR-related knowledge, ensuring it is curated, classified, and readily accessible to all stakeholders.

Activity 2.1: Identify, collect, and digitize existing FLR training materials, manuals, case studies, learning modules, and technical reports from national and international sources.

A systematic approach will be employed to gather, digitize, and prepare a rich repository of FLR knowledge.

* **Inventory of National and International Sources:** We will compile a comprehensive inventory of national and international sources, including MoEFCC, State Forest Departments, ICFRE, ICIMOD, FAO, IUCN, CIFOR-ICRAF, UNCCD, and relevant academic institutions, for FLR-related knowledge materials.
* **Engagement for Grey Literature:** We will actively engage with consortium partners and stakeholders to identify unpublished or grey literature, internal training modules, and field manuals that are highly relevant to FLR but not widely accessible.
* **Standardized Collection Framework:** A standardized collection framework will be established to categorize resources consistently by type (manual, case study, training module, technical report, policy note) and theme (agroforestry, community forestry, financing, monitoring).
* **Copyright and Licensing Verification:** Permissions will be requested, and copyright/licensing status will be meticulously verified for all collected materials to ensure full compliance with intellectual property laws and open-access requirements.
* **Format Conversion to Web-Compatible Formats:** Physical or legacy-format materials (PDF, Word,

PowerPoint, video, audio, images) will be collected from institutions and converted into digital, web-compatible formats such such as HTML5, tagged PDF, or embedded media for optimal portal integration.
* **Bilingual Processing and Localization:** Key resources will undergo bilingual processing, including translation into Hindi where necessary, and localization to ensure cultural and contextual relevance for the target audience.
* **OCR for Searchability and Accessibility:** Optical Character Recognition (OCR) tools will be applied to all scanned documents to ensure the text is fully searchable and compliant with accessibility standards.
* **Metadata Field Establishment:** Essential metadata fields will be established for each resource, including title, author, year, theme, geography, target audience, and keywords, to support structured cataloguing and discovery.
* **Centralized Repository with Version Control:** Digitized resources will be stored in a centralized, secure repository with robust version control and regular backup mechanisms to prevent data loss and manage content evolution.
* **Validation for Completeness and Accessibility:** All digitized outputs will be rigorously validated for completeness, readability, and compliance with WCAG 2.1 standards before being uploaded into the portal.
* **Quality Assurance Checks:** Comprehensive quality assurance checks will be conducted to confirm the accuracy of translations, consistency of formatting, and integrity of all multimedia resources.
* **Digitization Log:** A detailed digitization log will be maintained, tracking the source, date of collection, format converted, and responsible staff for each resource, ensuring accountability and traceability.
* **Stakeholder Review of Sample Materials:** Stakeholders will be engaged in reviewing sample digitized materials to confirm their usability and alignment with their training and knowledge needs.
* **Integration into CMS with Metadata Tagging:** Finalized resources will be seamlessly integrated into the portal?s content management system, with appropriate metadata tagging applied for easy discovery and retrieval through the portal's search and navigation features.

Activity 2.2: Review content to assess quality, relevance, and licencing requirements (open access, copyright).

A systematic content review process will ensure that all integrated resources meet high standards of quality, relevance, and legal compliance.

* **Content Review Framework Development:** A comprehensive content review framework will be developed, outlining clear criteria for assessing quality, relevance, and compliance with licensing requirements.
* **Systematic Resource Classification:** All collected resources will be systematically classified into predefined categories to facilitate an organized and efficient review process.
* **Technical Accuracy Assessment:** Technical accuracy will be assessed by cross-referencing information with current FLR practices, policies, and scientific literature to ensure the credibility and reliability of the content.
* **Contextual Relevance Evaluation:** The contextual relevance of each resource will be evaluated to ensure it aligns with the specific needs, challenges, and priorities of the RECAP4NDC project and its target stakeholder groups in India and South Asia.